METRIC

MIL-DTL-63245C(AR)
w/AMENDMENT 2
29 June 2017
SUPERSEDING
MIL-DTL-63245C
w/ AMENDMENT 1
20 April 2016

DETAIL SPECIFICATION

PROJECTILE 155 MILLIMETER: HE, METAL PARTS ASSEMBLY, M795

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center(ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for the manufacturing and assembly of the metal parts for Projectile, HE, 155MM, M795 (see 6.1).

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to assure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	- Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	- Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	- DoD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, Copies of these documents are available online at http://quicksearch.dla.mil/ from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

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US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

9280308 - Ultrasonic Inspection Standard, Longitudinal and Radial Cracks, M795, Body

9280309 - Ultrasonic Inspection Standard Circumferential and Piping Defects 155MM, M795

13066810 - Ultrasonic Testing Standard, Wall Thickness and Variation, M795 Body

9280336 - Envelope Drawing, Ultrasonic Inspection Equipment and Procedure for Body Machined, 155MM, M795

9280429 - Rotating Band Seat Tester.

9280441 - Envelope Drawing, Magnetic Particle Inspection, Band Area, 155MM, M795

9326052 - Blank Rotating Band

9326053 - Body Machined

9326054 - Body Assembly

9326055 - Projectile, 155MM, HE, M795 Metal Parts Assembly

12977199 - Body Machined

12977200 - Body Assembly

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AEROSPACE INDUSTRIES ASSOCIATION of AMERICA, Inc. (AIA/NAS)

NAS 410 – NAS Certification & Qualification of Nondestructive Test Personnel

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc, (AIA/NAS), 1250 Eye Street, NW; Suite 1100 Washington, DC 20005.)

ASTM INTERNATIONAL (ASTM)

ASTM B 117 - Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM E 8 - Standard Test Method for Tension Testing of Metallic Materials
ASTM E 10 - Standard Test Method for Brinell hardness of Metallic Materials
ASTM E 18 - Standard Test Method for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E 103 - Standard Test Method for Rapid Indentation Hardness Testing of Metallic Materials
ASTM E 399 – Standard Test Method for Plane-Strain Fracture Toughness of Metallic Materials
ASTM E 1444– Standard Practice for Magnetic Particle Testing

(Copies of ASTM standards are available online at http://www.astm.org or from American Society for Testing and materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9326055. Part number 9326055-1 utilizes body assembly 12977200 with a welded rotating band and Part number 9326055-2 utilizes body assembly 9326054 with a swaged rotating band.

3.2.1 Blank, Rotating Band. (Swaged band)(if applicable) The materials, components and assemblies shall be in accordance with drawing 9326052.

3.2.2 Body machined (mult). (Swaged band)(if applicable) The materials, components and assemblies shall be in accordance with drawing 9326053.

3.2.3 Body machined. (prior to nosing)(Swaged band)(if applicable) The materials, components and assemblies shall be in accordance with drawing 9326053.

3.2.4 Body machined. (Swaged band)(if applicable) The materials, components and assemblies shall be in accordance with drawing 9326053.

3.2.5 Body Assembly. (Swaged band)(if applicable) The materials, components and assemblies shall be in accordance with drawing 9326054.

3.2.6 Body machined (mult). (Welded band) The materials, components and assemblies shall be in accordance with drawing 12977199.

3.2.7 Body machined (prior to nosing). (Welded band) The materials, components and assemblies shall be in accordance with drawing 12977199.

3.2.8 Body Machined. (Welded band) The materials, components and assemblies shall be in accordance with drawing 12977199.

3.2.9 Body Assembly. (Welded band) The materials, components and assemblies shall be in accordance with drawing 12977200.

3.2.10 Projectile, HE, 155MM, M795 Metal Parts Assy. The materials, components and assemblies shall be in accordance with drawing 9326055.

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3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Mechanical properties. Heat treatment or stress relief applied shall be uniform throughout each lot of projectiles and shall be so performed and controlled as to produce in the projectile the mechanical properties specified in 9326053 or 12977200 as applicable.

3.5 Plane-Strain Fracture Toughness test. The minimum fracture toughness (K_{ic}) in the side wall shall be 25 KSI$\sqrt{in}$. (thousand pounds per square inch square root inch). The minimum fracture toughness (K_{ic}) in the base shall be 22.5 KSI$\sqrt{in}$. Independent verification of fracture toughness shall be conducted in accordance with this specification.

3.6 Projectile material defects.

3.6.1 Projectile body. The projectile body shall be free from cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects. Visual standards for evaluation of metal defects will be established for each contract after the first month of production. However, visual standards shall not preclude the requirement for compliance with 3.11 (see 6.4).

3.6.2 Projectile cavity. In addition to the requirements of 3.6.1 the projectile cavity shall be free from scale, fins, burrs, draw marks, laminations, imbedded foreign matter and pits. Surface discontinuities which have a radius smaller than 0.76 mm (.03 inch) shall be rejected. Visual standards for evaluation of metal defects will be established for each contract after the first month of production. However, visual standards shall not preclude the requirement for compliance with 3.11 (see 6.4).

3.7 Swaged rotating band. (If applicable)

3.7.1 Band seat. Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust, or other foreign matter, Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.7.2 Rotating band mechanical properties. After application and subsequent removal the rotating band shall have the following mechanical properties:

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a. Tensile Strength	–	206.8 Megapascals (MPa) (30,000 lbs per square inch (PSI)), minimum
b. Elongation	–	25 percent in two centimeters of gage length, minimum

3.7.3 Band seat sidewall contact. The rotating band shall be applied after final heat treatment of the body, by carefully pressing the band blank into the band seat so as to make contact with the sidewalls for a minimum of eight-tenths of the band seat depth. The band shall be applied in such a manner as not to distort the body wall. In addition, There shall be no surface gap between the rotating band and the sidewalls of the band seat, after the rotating band has been machined to final dimensions.

3.7.4 Band seating tightness. The diametrical clearance between the rotating band and the band seat shall not exceed .23 millimeters (.009 inches) when the non-destructive method of testing is used. The diametrical clearance between the rotating band and the band seat shall not exceed .15 millimeters (.006 inches) when the destructive method of testing is used.

3.8 Welded rotating band.

3.8.1 Foreign matter on band seat. Prior to banding, the band seat shall be free from any contaminants that would be detrimental to the adhesion of the band to the band seat.

3.8.2 Rotating band seat control. Body material (steel) under the rotating band shall not be visible in the cannelure diameter. Body material (steel) under the rotating band shall not extend above the finished bourrelet surface.

3.8.3 Minimum melting of steel. Upon removal of the rotating band material, the steel band seat shall demonstrate visual evidence of minimum melting due to gilding metal adhesion during application of the rotating band. Visual comparison standards will be provided by the Government.

3.8.4 Band seat subsurface defects. There shall be no evidence of intergranular penetration of gilding metal, weld arc cratering except for the starting weld pass or cracks when the body band seat is turned down 147.19 millimeters (mm) plus or minus 0.25 mm (5.795 inches plus or minus .010 inches) diameter

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and tested as specified in 4.4.5.3. There shall be no evidence of intergranular penetration of gilding metal, weld arc cratering or cracks when the body band seat is turned down to 144.65 mm plus or minus 0.25 mm (5.695 inches plus or minus .010 inches) diameter and tested as specified in 4.4.5.3.

3.8.5 Band welding stress relief. The projectile shall be quench and temper heat treated or stress relieved within 4 hours of overlay band welding. Stress relief heat treatment shall be at a minimum of 538 degrees C (1000 degrees F) for a minimum of 1 hour. If induction is employed for stress relieving the maximum hardness of the heat affected zone after tempering shall be Rockwell C40. The peak temperature at any point in the steel body shall not exceed the lower austenite transformation temperature or cause melting of the gilding metal.

3.8.6 Rotating band surface defects. The surface of the rotating band shall be examined for surface defects, damage and contamination. Visual standards for evaluation of rotating band surface defects will be established for each contract after the first month of production.

a. On the forward land, there shall be no evidence of raised copper due to nicks, digs or damage of any kind and there shall be no paint or foreign matter on the outer diameter.

b. On the aft three lands, any nicks or digs extending 1/2 the land width or less are acceptable, provided the maximum band diameter is not exceeded.

c. On the surface of the overlay band the number of cavities equal to or less than 3.97 mm (5/32 inch) in the greatest dimension shall not exceed one (1) per 45 degree segment of the band. The allowable cavity on the forward or aft bands shall not exceed 2.38 mm (3/32 inch) in the greatest dimension. Any cavity greater than 2.38 mm (3/32 inch) must be filled with Government approved filler/hardener. However one (1) 654.16 square mm (1 square inch) area per band is allowed a maximum of four (4) cavities, none of which exceeds 2.38 mm (3/32 inch) in the largest dimension.

d. One tightly closed seam in the area of the weld start stop zone is allowed on the surface of the overlay band.

e. Disbond (pick out) as indicated by a ragged appearance forward or aft of the rotating band is allowed.

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f. The overlay rotating band and band seat areas shall have no other discontinuities, cracks, lack of bonding, lack of fill or other metal defects other than those herein defined.

3.9 Protection of rotating band. The rotating band shall be protected after final machining from being nicked, burred, or otherwise damaged in subsequent handling prior to shipment.

3.10 Hydrostatic test. The projectile shall withstand an internal hydrostatic pressure in accordance with drawing 9326053 or 12977200 as required without evidence of leakage, cracks, ruptures or permanent distortion.

3.11 Ultrasonic test. Projectiles shall be free of any discontinuities equal to or in excess of the ultrasonic inspection criteria for the respective zones. The inspection criteria are notches and holes in the standards (see 9280308 and 9280309), which simulate cracks and piping defects. Additionally, projectiles shall be free of wall variation and undersized wall thickness equal to or in excess of the ultrasonic inspection criteria for the respective zones. The inspection criteria are holes and machined bands in the standard (See 13066810), which defines wall thickness and variation.

3.12 Interior cavity finish. The interior cavity after painting shall be free from bare spots, blisters, grease, dirt and other foreign matter. There shall be no pool of paint in the bottom of the cavity.

3.13 Painting. All paint shall be dry to the touch before packing for shipment. The paint shall be considered dry when the soft tacky condition no longer exists and the paint feels firm when touched with a finger using moderate pressure.

3.14 Ballistics test. The projectile assembly shall reveal no evidence of metal parts failure. Metal parts failure shall be defined as:

1. Breakup. Loss or breakup of metal parts components in the gun bore or in flight.

2. Rotating band failure. Loss or fanning of the rotating band attributable to noncompliance with drawing 9326054 or 12977200 as required.

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3. Shearing of rotating band. Shear of rotating band as evidenced by complete lack of engraving.

3.15 Workmanship. All parts and assemblies shall be fabricated and assembled in a thorough, skillful manner. Components shall be free of burrs, sharp edges, cracks, dirt, grease, corrosion, and other foreign matter. Cleaning methods used shall not be injurious to any parts, nor shall the cleaning agents contaminate the parts. All required marking and stamping shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Methods of Verification N/A – Not Applicable 1 – Analysis 3 – Examination 2 – Demonstration 4 – Test							Classes of Verification A- First Article B- Acceptance		
Section 3 Req.	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.1	First Article Test				X	X	X	X	4.2
3.2	Components and assemblies	X							
3.2.1	Blank, Rotating Band				X	X	X	X	4.3.2.1
3.2.2	Body machined (mult)				X	X	X	X	4.3.2.2
3.2.3	Body machined (prior to nosing)				X	X	X	X	4.3.2.3
3.2.4	Body machined				X	X	X	X	4.3.2.4
3.2.5	Body Assembly				X	X	X	X	4.3.2.5
3.2.6	Body machined (mult)				X	X	X	X	4.3.2.6
3.2.7	Body machined (prior to nosing)				X	X	X	X	4.3.2.7
3.2.8	Body Machined				X	X	X	X	4.3.2.8
3.2.9	Body Assembly				X	X	X	X	4.3.2.9
3.2.10	Projectile, HE, 155MM,M795 Metal Parts Assembly				X	X	X	X	4.3.2.10
3.3	Ammunition lot numbering				X		X	X	4.3.2.10
3.4	Mechanical properties					X	X	X	4.4.3
3.5	Plane-strain fracture toughness test					X	X	X	4.4.3.3
3.6	Projectile material defects			X			X	X	4.3.2.4 4.3.2.8 & 4.3.2.9
3.6.1	Projectile body			X			X	X	4.3.2.4, 4.3.2.8 & 4.3.2.9
3.6.2	Projectile cavity			X			X	X	4.3.2.4 4.3.2.8 & 4.3.2.9
3.7	Swaged rotating band			X	X		X	X	4.4.4
3.7.1	Band seat			X	X		X	X	4.3.2.3

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TABLE I. Requirement/verification cross reference matrix - Continued

Section 3 Req.	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.7.2	Rotating band mechanical properties			X	X		X	X	4.4.4.3
3.7.3	Band seat sidewall contact			X	X		X	X	4.4.4.1
3.7.4	Band seating tightness			X	X		X	X	4.4.4.1 & 4.4.4.2
3.8	Welded rotating band			X	X		X	X	4.4.5
3.8.1	Foreign matter on band seat			X	X		X	X	4.3.2.7
3.8.2	Rotating band seat control			X	X		X	X	4.4.5.1
3.8.3	Minimum melting of steel			X	X		X	X	4.4.5.1
3.8.4	Band seat subsurface defects			X	X		X	X	4.4.5.1
3.8.5	Band welding stress relief			X	X		X	X	4.4.5.4
3.8.6	Rotating band surface defects			X	X		X	X	4.3.2.9
3.9	Protection of rotating band			X	X		X	X	4.3.2.9
3.10	Hydrostatic test			X	X		X	X	4.4.6
3.11	Ultrasonic test			X	X		X	X	4.4.7
3.12	Interior cavity finish			X	X		X	X	4.3.2.9
3.13	Painting			X	X		X	X	4.3.2.9
3.14	Ballistics test			X	X		X	X	4.4.10
3.15	Workmanship			X	X		X	X	4.3.2.1-4.3.2.9

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4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table II.

4.2.2 Inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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TABLE II. First article inspection

TITLE Projectile assemblies and components	**SHEET** 1 **OF** 2	**DRAWING NUMBER** See below

EXAMINATION OR TEST	**NUMBER OF SAMPLES**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Blank rotating band drawing. 9326052 (if applicable)	(25)	3.2.1	4.3.2.1
Examination for defects	25	3.2.1	4.4.4.3
Mechanical properties	5	3.2.2	4.3.2.2
Body, machined drawing. 9326053 (if applicable)	(25)	3.2.3	4.3.2.3
Maximum temperature of mult (Note 1)	Note 1	3.2.3	4.3.2.3
Examination for defects (prior to nosing)	25	3.10	4.4.6
Examination for defects	25	3.11	4.4.7
Hydrostatic test	25	4.4.3.1	4.4.3.1.1
Ultrasonic inspection	25	3.2.4	Certification
Hardness test	25	3.4	4.4.3.2.2
Chemical composition (Note 2)	1	3.5	4.4.3.3
Mechanical properties test (Note 2)	5	3.7.2	4.4.4
Plane-Strain Fracture Toughness test (Note 2)	1	3.7.4	4.4.4.1
Rotating band mechanical properties	5	3.7.3	4.4.4.1
Rotating band diametrical clearance	5	3.2.4	4.3.2.4
Rotating band side wall contact	5		
Body assembly drawing.9326054 (if applicable)	(25)		
Examination for defects	25		

Note 1: Check temperature of first, middle and last mult to be used in the first article sample.
Note 2: Verification by an independent lab is required.

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TABLE II. First article inspection continued

TITLE Projectile assemblies and components	**SHEET** 2 **OF** 2	**DRAWING NUMBER** See Below	
EXAMINATION OR TEST	**NUMBER OF SAMPLES**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Body, machined drawing. 12977199	(25)		
Maximum temperature of mult (Note 1)	3 (Note 3)	3.2.6	4.3.2.6
Examination for defects (prior to nosing)	25	3.2.7	4.3.2.7
Examination for defects	25	3.2.8	4.3.2.8
Chemical composition (Note 2)	1	3.2.6	Certification
Body assembly drawing. 12977200	(25)		
Examination for defects	25	3.2.9	4.3.2.9
Hydrostatic test	25	3.10	4.4.6
Ultrasonic inspection	25	3.11	4.4.7
Magnetic particle test	5*	3.8.4	4.4.5.3
Hardness test	25	3.4	4.4.3.1
Mechanical properties test (Note 2)	2*	3.4	4.4.3.2
Plane-strain fracture toughness test (Note 2)	1*	3.5	4.4.3.3
Band chemical analysis	5*	3.2.9	4.4.5.1
Band welding stress relief (if applicable)	2*	3.8.5	4.4.5.4
Projectile, 155M, HE, M795 Metal Parts	(45)		
Assembly drawing. 9326055			
Examination for defects	45	3.2.10	4.3.2.10
Salt Spray Test	5	3.2.10	4.4.9
Ballistic performance for structural integrity	20	3.14	4.4.10

Note 1: Check temperature of first, middle and last mult to be used in the first article sample.
Note 2: Verification by an independent lab is required.
Note 3: This reading is to be taken during processing of First Article samples.
*These samples are in addition to samples used for examination of characteristics.

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916.

4.3.1.1 Heat treatment inspection lot. Heat treatment inspection lots shall not exceed one mill heat of steel or 2500 components, whichever is smallest. Whenever a break in heat treating of eight (8) hours or more occurs or a furnace is shut down for any length of time, the heat treatment inspection lot will be terminated and a new lot shall start. Additionally, heat treatment inspection lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment inspection lots shall remain grouped together and identified by inspection lot until mechanical property tests have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots.

4.3.2 Classification of characteristics. Conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. When cited herein, attributes sampling inspection shall be conducted in accordance with MIL-STD-1916, using the inspection levels cited in the Classification of Characteristic paragraphs. Definitions of Critical, Major and Minor shall be as defined in MIL-STD-1916. Acceptance criteria shall be in accordance with MIL-STD-1916. Alternative conformance inspections may be submitted and approved in accordance with MIL-STD-1916. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided there is an established correlation between the tool dimensions and the component dimensions prior to the start of production. There shall be adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1	Blank, rotating band (swaged band) (if applicable)			9326052
				NEXT HIGHER ASSEMBLY 9326054
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Inside diameter	Level IV	3.2.1	AIE
102	Width	Level IV	3.2.1	AIE
103	Thickness	Level IV	3.2.1	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.15	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.3.2.2	Body machined (mult) (swaged band) (if applicable)			9326053
				NEXT HIGHER ASSEMBLY 9326054
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Chemical composition certification (Note 1)	Lot of steel	3.2.2	Certification
102	Temperature of mult, max.	One per hour	3.2.2	4.4.1
Minor	None defined			

Note 1. Independent verification of chemical composition. Beginning with the first heat of material and once per year thereafter as a minimum a sample of the material being used for production of the projectile body shall be sent to a certified lab for independent analysis of chemical composition. If the independent analysis indicates that any sample fails to meet any requirement for correct chemical composition of the alloy, all material represented by the sample and all applicable components made since the last successful analysis shall be rejected.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Body machined (prior to nosing) (swaged band)		SHEET 1 OF 1	DRAWING NUMBER 12977199 NEXT HIGHER ASSEMBLY 12977200
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Presence of metal defects projectile cavity	100%	3.6.2	4.4.2/Visual/dedicated light source
Minor	None defined			

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		SHEET 1 OF 3	DRAWING NUMBER
4.3.2.4	Body machined (swaged band) (if applicable)			9326053
				NEXT HIGHER ASSEMBLY 9326054
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Hydrostatic test	100%	3.10	4.4.6
2	Ultrasonic test	100%	3.11	4.4.7
Major				
101	Hardness test	100%	4.4.3.1	4.4.3.1.1
102	Yield strength and elongation of body	4.4.3.2.1	3.4	4.4.3.2.2
103	Plane-Strain Fracture Toughness test	4.4.3.3	3.5	4.4.3.3
104	Diameter of rotating band seat	Level IV	3.2.4	AIE
105	Runout of rotating band seat	Level IV	3.2.4	AIE
106	Location of rotating band seat	Level IV	3.2.4	AIE
107	Width of rotating band seat	Level IV	3.2.4	AIE
108	Thickness through the base	Level IV	3.2.4	AIE
109	Scoring or knurling missing from rotating band seat	Level IV	3.2.4	Visual
110	Depth of scoring	Level IV	3.2.4	AIE
111	Foreign matter on rotating band seat	Level IV	3.2.4	Visual
112	Runout of pitch diameter of thread with bourrelet and front face	Level IV	3.2.4	AIE
113	Runout of ogive with bourrelet (three locations)	Level IV	3.2.4	AIE
114	Runout of bourrelet relief with bourrelet	Level IV	3.2.4	AIE
115	Runout of boattail with bourrelet	Level IV	3.2.4	AIE

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4 Cont.	TITLE Body machined (swaged band) (if applicable)		SHEET 2 OF 3	DRAWING NUMBER 9326053 / NEXT HIGHER ASSEMBLY 9326054
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
116	Diameter of bourrelet, min.	Level IV	3.2.4	AIE
117	Diameter of bourrelet relief	Level IV	3.2.4	AIE
118	Length from front face to first basic diameter	Level IV	3.2.4	AIE
119	Length from front face to second basic diameter	Level IV	3.2.4	AIE
120	Diameter at 186.34mm from forward end	Level IV	3.2.4	AIE
121	Diameter at 313mm from forward end	Level IV	3.2.4	AIE
122	Length from rear face to forward basic diameter on boattail	Level IV	3.2.4	AIE
123	Length from rear face to rear basic diameter on boattail	Level IV	3.2.4	AIE
124	Length from front face to relief of ogive	Level IV	3.2.4	AIE
125	Length from rear face to bottom of cavity	Level IV	3.2.4	AIE
126	Length from front face to rear of bourrelet relief	Level IV	3.2.4	AIE
127	Length from front face to forward face of bourrelet relief	Level IV	3.2.4	AIE
128	Thickness of body wall in ogive area, min. (2 locations)	100%	3.2.4	4.4.7.1/Note A
129	Variation in wall thickness in ogive area	100%	3.2.4	4.4.7.1/Note A
130	Variation in wall thickness in bourrelet relief	100%	3.2.4	4.4.7.1/Note A
131	Pitch diameter of thread, max.	Level IV	3.2.4	AIE
132	Minor diameter of thread, max	Level IV	3.2.4	AIE
133	Length of effective thread, min	Level IV	3.2.4	AIE

Note A – Wall variation will be checked across the entire length of the ogive area and bourrelet relief (See 12977199, Note 12).

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4 Cont.	TITLE Body Machined (swaged band) (if applicable)		SHEET 3 OF 3	DRAWING NUMBER 9326053 / NEXT HIGHER ASSEMBLY 9326054
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
134	Foreign matter in cavity	Level IV	3.2.4	Visual
135	Length total	Level IV	3.2.4	AIE
136	Runout of boattail cavity	Level IV	3.2.4	AIE
137	Runout aft of band seat	Level IV	3.2.4	AIE
138	Diameter aft of band seat	Level IV	3.2.4	AIE
Minor				
201	Diameter of thread chamfer	Level II	3.2.4	AIE
202	Diameter of rear of boattail (I.D.)	Level II	3.2.4	AIE
203	Evidence of shot or sandblasting on rotating band seat	Level II	3.2.4	Visual
204	Pitch diameter of thread, Min	Level II	3.2.4	AIE
205	Radii of chamfers missing or incorrect	Level II	3.2.4	Visual
206	Evidence of poor workmanship	Level II	3.15	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5	TITLE Body Assembly (swaged band) (if applicable)		SHEET 1 OF 2	DRAWING NUMBER 9326054 / NEXT HIGHER ASSEMBLY 9326055
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Presence of metal defects projectile cavity	100%	3.6.2	4.4.2/Visual/ dedicated light source
102	Diameter of rotating band ribs	Level IV	3.2.5	AIE
103	Runout of rotating band with bourrelet	Level IV	3.2.5	AIE
104	Diameter of rotating band seat	Level IV	3.2.5	AIE
105	Width of obturator band seat	Level IV	3.2.5	AIE
106	Location of cannelure (3 locations)	Level IV	3.2.5	AIE
107	Diameter of rotating band cannelure (3 locations)	Level IV	3.2.5	AIE
108	Location of rotating band	Level IV	3.2.5	AIE
109	Band seating tightness	Level IV	3.7.5	4.4.4
110	Rotating band mechanical properties	Level IV	3.7.3	4.4.4.3
111	Rotating band side wall contact	Level IV	3.7.4	4.4.4/Visual
112	Weight	Level IV	3.2.5	AIE
113	Foreign matter in band seat	Level IV	3.2.5	Visual
114	Location of basic diameter (Rotating band)	Level IV	3.2.5	AIE

- Prior to banding
- 1. Any part found without the required test stamp shall be tested to determine if the part fails the critical test. If the part fails the critical test it shall be treated as a Critical Nonconformance. If the part passes the critical test it shall be treated as a Major Nonconformance.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5 Cont.	TITLE Body Assembly (swaged band) (if applicable)		SHEET 2 OF 2	DRAWING NUMBER 9326054 / NEXT HIGHER ASSEMBLY 9326055
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
115	Presence of metal defects projectile body	100%	3.6.1	Visual
116	Hydrostatic test stamp missing Note 1	100%	3.2.5	Visual
117	Ultrasonic test stamp missing Note 1	100%	3.2.5	Visual
Minor				
201	Width of undercuts forward and rear of Rotating band (if applicable)	Level II	3.2.5	AIE
202	Depth of undercuts forward and rear of rotating band (if applicable)	Level II	3.2.5	AIE
203	Length, total, max.	Level II	3.2.5	AIE
204	Stamping missing, misleading or unidentifiable	Level II	3.2.5	Visual
205	Radii or chamfers missing or incorrect	Level II	3.2.5	Visual
206	Evidence of poor workmanship	Level II	3.15	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.3.2.6	Body machined (mult) (welded band)			12977199
				NEXT HIGHER ASSEMBLY
				12977199
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Chemical composition certification Note 1	Lot of steel	3.2.2	Certification
102	Temperature of mult, max.	One per hour	3.2.6	4.4.1
Minor	None defined			

Note 1. Independent verification of chemical composition. Beginning with the first heat of material and once per year thereafter as a minimum a sample of the material being used for production of the projectile body shall be sent to a certified lab for independent analysis of chemical composition. If the independent analysis indicates that any sample fails to meet any requirement for correct chemical composition of the alloy, all material represented by the sample and all applicable components made since the last successful analysis shall be rejected.

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.3.2.7	Body machined (prior to nosing) (welded band)			12977199
				NEXT HIGHER ASSEMBLY
				12977200
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Presence of metal defects projectile cavity	100%	3.6.2	4.4.2/Visual/ dedicated light source
Minor	None defined			

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.8	TITLE Body machined (welded band)		SHEET 1 OF 3	DRAWING NUMBER 12977199 / NEXT HIGHER ASSEMBLY 12977200
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Cleanliness of rotating band seat (*)	Level IV	3.2.8	Visual
102	Diameter of rotating band seat (*)	Level IV	3.2.8	AIE
103	Runout of rotating band seat (*)	Level IV	3.2.8	AIE
104	Location of rotating band seat (*)	Level IV	3.2.8	AIE
105	Width of rotating band seat (*)	Level IV	3.2.8	AIE
106	Thickness through the base	Level IV	3.2.8	AIE
107	Runout of pitch diameter of thread with bourrelet and front face	Level IV	3.2.8	AIE
108	Runout of ogive with bourrelet (three Locations)	Level IV	3.2.8	AIE
109	Runout of bourrelet relief with bourrelet	Level IV	3.2.8	AIE
110	Runout of boat tail with bourrelet	Level IV	3.2.8	AIE
111	Diameter of bourrelet, min	Level IV	3.2.8	AIE
112	Diameter of bourrelet relief	Level IV	3.2.8	AIE
113	Length from front face to first basic diameter	Level IV	3.2.8	AIE
114	Length from front face to second basic diameter	Level IV	3.2.8	AIE
115	Diameter of ogive at long basic length	Level IV	3.2.8	AIE
* Prior to banding				

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.8 Cont.	TITLE Body machined (welded band)		SHEET 2 OF 3	DRAWING NUMBER 12977199 / NEXT HIGHER ASSEMBLY 12977200
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
116	Diameter of ogive at shorter basic length	Level IV	3.2.8	AIE
117	Length from rear face to forward basic diameter on boattail	Level IV	3.2.8	AIE
118	Length from rear face to rear basic diameter on boattail	Level IV	3.2.8	AIE
119	Length from front face to relief of ogive	Level IV	3.2.8	AIE
120	Length from rear face to bottom of cavity	Level IV	3.2.8	AIE
121	Length from front face to rear of bourrelet relief	Level IV	3.2.8	AIE
122	Length from front face to forward face of bourrelet relief	Level IV	3.2.8	AIE
123	Thickness of body wall in ogive area, min. (2 location)	100%	3.2.8	4.4.7.1/Note A
124	Variation of wall thickness in ogive area	100%	3.2.8	4.4.7.1/Note A
125	Variation of wall thickness in bourrelet relief	100%	3.2.8	4.4.7.1/Note A
126	Pitch diameter of threads, max.	Level IV	3.2.8	AIE

Note A – Wall variation will be checked across the entire length of the ogive area and bourrelet relief (See 12977199, Note 12).

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.8 Cont.	TITLE Body machined (welded band)		SHEET 3 OF 3	DRAWING NUMBER 12977199 / NEXT HIGHER ASSEMBLY 12977200
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
127	Minor diameter of thread, max.	Level IV	3.2.8	AIE
128	Length of effective thread, min.	Level IV	3.2.8	AIE
129	Length, total	Level IV	3.2.8	AIE
130	Runout of boattail cavity	Level IV	3.2.8	AIE
131	Runout aft of band seat	Level IV	3.2.8	AIE
132	Diameter aft of band seat	Level IV	3.2.8	AIE
Minor				
201	Radii or chamfers missing	Level II	3.2.8	Visual
202	Surface finish improper	Level II	3.2.8	Visual
203	Foreign matter in cavity	Level II	3.2.8	Visual
204	Evidence of poor workmanship	Level II	3.15	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.9	TITLE Body assembly (welded band)		SHEET 1 OF 2	DRAWING NUMBER 12977200 / NEXT HIGHER ASSEMBLY 9326055
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Hydrostatic test	100%	3.10	4.4.6
2	Ultrasonic test	100%	3.11	4.4.7
3	Magnetic particle test (band area)	4.4.5.3	3.8.4	4.4.5.3
Major				
101	Hardness test	100%	4.4.3.1	4.4.3.1.1
102	Yield strength and elongation of body Note 1.	4.4.3.2	3.4	4.4.3.2
103	Plane-Strain Fracture Toughness test Note 1.	4.4.3.3	3.5	4.4.3.3
104	Properties of rotating band weld	4.4.5.1	3.8.3/ 3.8.4	4.4.5.1
105	Rotating band chemical analysis	4.4.5.1	3.2.9	4.4.5.1 a
106	Diameter of rotating band	Level IV	3.2.9	AIE
107	Runout of rotating band with bourrelet	Level IV	3.2.9	AIE
108	Diameter of obturator band seat	Level IV	3.2.9	AIE
109	Width of obturator band seat	Level IV	3.2.9	AIE
110	Location of cannelure (3 locations)	Level IV	3.2.9	AIE
111	Diameter of rotating band cannelure (3 locations)	Level IV	3.2.9	AIE
112	Location of rotating band	Level IV	3.2.9	AIE

Note 1. Independent verification of this characteristic is required. Beginning with the first heat of material and once per year thereafter as a minimum a sample of the material being used shall be sent to a certified lab for independent analysis. If the independent analysis indicates that any sample fails to meet any requirement, all material represented by the sample and all applicable components made since the last successful analysis shall be rejected.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.9 Cont.	TITLE Body assembly (welded band)		SHEET 2 OF 2	DRAWING NUMBER 12977200 / NEXT HIGHER ASSEMBLY 9326055
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
113	Location of basic diameter (rotating band)	Level IV	3.2.9	AIE
114	Rotating band surface defective	Level IV	3.8.6	Visual
115	Length, total, max.	Level IV	3.2.9	AIE
116	Weight	Level IV	3.2.9	Visual/Probe/ Scale
117	Presence of metal defects	100%	3.2.9	Visual
118	Hydrostatic test stamp missing Note 2	100%	3.2.9	Visual
119	Ultrasonic test stamp missing Note 2	100%	3.2.9	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.15	Visual

2. Any part found without the required test stamp shall be tested to determine if the part fails the critical test. If the part fails the critical test it shall be treated as a Critical Nonconformance. If the part passes the critical test it shall be treated as a Major Nonconformance.

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.3.2.10	Projectile, HE, 155MM, M795 Metal Parts Assy.		1 OF 1	9326055
				NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Ballistic test failure, projectile break-up or component separation in bore or in flight	20/10	3.14	4.4.10
2	Ballistic test failure, rotating band failure	4.4.10	3.14	4.4.10
3	Ballistic test failure, shearing of rotating band	4.4.10	3.14	4.4.10
Major				
101	Diameter of bourrelet, max.	100%	3.2.10	Ring Gage
102	Rotating band damaged	100%	3.2. 10	Visual
103	Cleanliness of cavity	100%	3.2. 10	Visual
104	Paint on threads	100%	3.2. 10	Visual
105	Pool of paint in cavity	100%	3.2. 10	Visual
106	Presence of metal defects	100%	3.2. 10	Visual
107	Salt spray test	4.4.9	3.2. 10	4.4.9
Minor				
201	Presence of burrs	Level II	3.2. 10	Visual
202	Protective coating thickness	Level II	3.2. 10	Visual
203	Flexible rotating band cover missing or miss installed	Level II	3.2. 10	Visual
204	Marking missing, incorrect or illegible	Level II	3.2. 10	Visual
205	Ammunition lot numbering incorrect	Level II	3.2. 10	Visual
206	Plug, shock attenuator missing	Level II	3.2. 10	Visual
207	Evidence of poor workmanship	Level II	3.15	Visual

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4.4 Testing.

4.4.1 Mult temperature. One mult shall be checked for maximum temperature during each hour, or fraction thereof, of furnace operation. The inspection point and method shall require approval as part of the AIE submittal. If any mult checked exceeds the applicable requirement, all mults in that heat treatment batch shall be rejected.

4.4.2 Visual examination for material defects. The internal cavity of the projectile shall be inspected visually prior to nosing using probe lighting to assure conformance with the applicable requirements. Visual standards in accordance with 3.6.2 shall be set up for use during this inspection. Any projectile failing to meet the applicable requirements of 3.6.2 shall be classed defective and removed from the lot.

4.4.3 Mechanical properties.

4.4.3.1 Hardness. Each final heat treated projectile from a heat treated lot shall be identified and subjected to a hardness test on the surface 12.7 mm (1/2 inch) forward of the rotating band. One (1) hardness reading shall be taken. The hardness reading obtained shall be used as the basis for selecting the tension test sample of paragraph 4.4.3.2.

4.4.3.1.1 Hardness test. The hardness test shall be conducted in accordance with ASTM E 103 and the following:

a. The surface of the body in the area designated to be hardness tested shall be prepared by machining to a sufficient depth, 0.38 mm (.015 inch) minimum, to remove scale and get below the decarburization zone. The preparation process to be used shall require approval as part of the AIE submittal. Manual grinding shall not be used. The surface preparation requirement may be eliminated if the following conditions are met:

(1) Lab results confirm that current production process produces a decarburized zone of 0.076 mm (.003 inch) maximum in the test area.

(2) A controlled atmosphere is maintained in the austenitizing furnace.

(3) Depth of decarburization shall be verified whenever plane-strain fracture toughness testing is

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required (see 4.4.3.3), but only for the first heat treatment lot.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM E 10 in lieu of production samples.

c. The hardness tester shall use a standard 10mm Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000kg. The amount of penetration of the indentor (10mm ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

d. One hardness reading shall be taken.

e. All hardness test measurements shall be automatically recorded.

4.4.3.2 Mechanical properties test. Two final heat treated projectiles from a heat treated inspection lot shall be selected for the tension test. The projectiles selected shall be those which have the highest and the lowest hardness reading from 4.4.3.1. From each of these projectiles three specimens shall be tested in accordance with 4.4.3.2.1. In the event of rejection, new samples of higher or lower hardness values or both, as applicable, may be selected and tested in order to establish an acceptable hardness range. Once this range has been established all projectiles whose hardness values fall outside the range shall be rejected or subjected to retemper or heat treat cycle if approved by the Contracting Officer. An acceptable hardness range is one which will provide acceptable yield and elongation results.

4.4.3.2.1 Yield strength and elongation specimens. The two projectiles selected for testing shall have three specimens taken from each and tested as follows:

a. The base section tensile specimen shall be taken transversely. Failure to meet the yield strength or elongation requirements shall require a retest from the remaining portion

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of the base. Results of the retest specimen and the first failing specimen shall be averaged to determine compliance with the applicable drawing requirement. Failure of the average results to meet the drawing requirement shall be cause for rejection of the heat treated inspection lot.

b. The two longitudinal specimens shall be taken 180^0 from the sidewall. Location shall be centered under the rotating band seat and in the middle of the wall thickness in that area. Failure of any specimen to comply with the applicable drawing requirement shall be cause for rejection of the heat treated inspection lot.

4.4.3.2.2 Yield strength and elongation testing. Testing shall be conducted in accordance with ASTM E8 with the largest substandard size round specimen possible.

4.4.3.2.3 Independent verification of mechanical properties. Once per month, two additional samples shall be taken from each of the highest and the lowest hardness reading samples selected in 4.4.3.2 and sent to a certified lab for independent analysis of mechanical properties. The samples shall be taken adjacent to the original samples. If the independent analysis indicates that any sample fails to meet either the minimum yield strength or percent elongation, all product represented by the independent sample shall be rejected.

4.4.3.3 Plane-Strain Fracture Toughness test. Beginning with the first heat lot produced and continuing until three consecutive heat treat lots have complied with the applicable minimum requirement specified, one final heat treated projectile from each heat treated lot shall be subjected to the Plane-Strain Fracture Toughness test. The projectile selected shall be of the highest hardness reading from 4.4.3.1 and may be the same as used in 4.4.3.2. Four specimens shall be tested. No individual fracture toughness from the side wall shall be less than the applicable requirement. No individual fracture toughness from the base shall be less than the applicable requirement. If one test is invalid, acceptance may be based on the fracture toughness of the valid sample. If any individual fracture toughness is below the minimum allowable, the heat treated lot represented shall be rejected. The rejected lot may be subjected to retemper and retest, for all mechanical properties, if approved by the contracting officer. After three consecutive heat treated lots have been accepted, the test shall be conducted once per year. The test shall be reinstituted,

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until three consecutive heat treatment lots have complied with the applicable requirements, if any of the following occur:

a. Steel is procured from a different producer, or from a substantially different process.

b. Forgings are procured from a different producer.

c. A substantial change is made in the forging process or configuration.

d. A change is made in heat treatment contractors.

e. A substantial change is made in the heat treatment process.

f. A change is made to the heat treatment metal parts configuration.

4.4.3.3.1 Plane-Strain fracture toughness samples. The two base specimens shall be taken from locations F1 and F2 on Figure 1. The test shall be conducted at a specimen temperature of -53.9 degrees C (-65 degrees F). The test shall be conducted in accordance with the procedure defined in ASTM E399. The notches shall be machined across the rear face of the base specimen. Locations F3 and F4 on Figure 1 shall be used for replacement specimens for locations F1 and F2 respectively. The two side wall specimens shall be taken from locations F5 and F6 on Figure 2. The test shall be conducted at a specimen temperature of -53.9 degrees C (-65 degrees F). The test shall be conducted in accordance with the procedure defined in ASTM E399. Orientation of the side wall samples shall be L-R with the notches machined across the outside diameter surface of the specimen. Locations 180 degrees from samples F5 and F6 shall be used for replacement specimens for locations F5 and F6 respectively.

4.4.3.3.2 Independent verification of Plane-Strain fracture toughness. Once per year per contract requirement, two additional samples shall be taken from each of the base and side wall samples selected in 4.4.3.3 and sent to a certified lab for independent analysis. The samples shall be taken adjacent to the original samples. If the independent analysis indicates that any sample fails to meet Plane-Strain Fracture Toughness, all product represented by the independent sample shall be rejected.

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4.4.4 Swaged rotating band.

4.4.4.1 Rotating band swage machine sampling. Two body assemblies (one sample) shall be selected from each 4 hour production or fraction thereof from each banding machine in operation to determine compliance with applicable requirements. Failure to comply shall be cause for rejection of the assemblies represented by the sample. A banding machine shall be considered qualified after 40 consecutive samples have been found acceptable. The first body assembly shall be sectioned by cutting through two planes normal to the longitudinal axis one centimeter forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for inspection for mechanical properties and the other section for inspection of sidewall contact. The sample provided for mechanical properties shall have the rotating band removed and carefully flattened so as to hold a minimum to any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having a two centimeters (one inch) gage length and a one centimeter (.5 inch) gage width, and grip ends of sufficient length to accommodate the tensile machine. The specimen shall be tested in accordance with the procedure prescribed by ASTM method E-8. The result shall meet the requirements of 3.7.3. The other section shall be sectioned into two other parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove the metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.7.4. The second body assembly shall be used to determine diametrical clearance. The body assembly selected should not be processed through the machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the

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band; the results obtained shall be subtracted from the previously determined diameters over the band. The result shall meet the requirements of 3.7.5.
Failure to comply with any of the requirements shall be cause for rejection of the assemblies represented by the failing sample.

4.4.4.2 Rotating band swage machine reduced sampling. Five body assemblies from each 1-hour production or fraction thereof per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.6.3 shall be cause for the rejection of the assemblies represented by the samples. The samples provided by 4.4.4.1 shall be tested utilizing the equipment and procedures described in drawing No. 9280429 or the requirements of 6.5.2. In the event of a non destructive testing failure or failure to meet the requirement of paragraph 3.7.3, rotating band swaging shall immediately cease on the banding machine which produced the failure. The part in question shall be destructively tested to verify the gap. All bands since the last inspection shall be non-destructively tested for acceptance. An investigation into the cause of the failure shall be launched. Once determined and addressed swaging shall resume with the first 10 projectiles inspected. If all ten samples are acceptable inspection will return to post reduced sampling. If any of the 10 samples fail then the requirements of this paragraph shall be instituted.

4.4.4.3 Mechanical properties of rotating band blank. The band shall be cut from front to rear and flattened, without hammering which may further cold work the materiel, and the largest obtainable flat type specimen shall be cut from the band, the length of which shall provide a one inch gage length, plus adequate material at each end to accommodate the grips of the tensile testing machine. This method should yield a .500 inch gage width. In any event, the gage cross section shall be determined to the nearest .001 square inch and utilized to calculate the tensile strength, and elongation. Testing shall be conducted in accordance with ASTM E8.

4.4.5 Welded rotating band

4.4.5.1 Rotating band welder sampling. The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to the test outlined below. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been found

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acceptable. A welding machine shall be considered qualified after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been found acceptable. These bodies shall have the band removed by machining and chemically etching without damaging the band seat. These bodies will be subjected to the following tests:

a. Band chemical analysis. Sample chips shall be obtained from each band from the last 1.02 mm (.04 inch) removed when machining to the finished diameter. All chips shall be collected and identified with the body from which the chips were machined. A minimum 5 gram (.176 ounce) sample from each complete specimen thus obtained from each band shall be analyzed by any method that has been approved by the technical agency. The sample may not be cleaned with a magnet. Failure to comply with the requirements of the drawing shall be cause for rejection of the production represented by the sample.

b. Visual examination for compliance with 3.8.3 minimum melt requirements. Failure to comply with the requirements of the drawing shall be cause for rejection of the production represented by the sample.

c. Visual examination for compliance with 3.8.4 metal defects, using the technique described in 4.4.5.3. Turn down the band seat to the largest diameter specified in 3.8.4 and visually inspect in accordance with 4.4.5.3. Any defect indication per 3.8.4 other than at the starting weld pass is considered to be a reject. If indications are present at the starting weld pass turn down the band seat to the smallest diameter specified in 3.8.4 and again visually inspect in accordance with 4.4.5.3. Any defect per 3.8.4 is considered to be a reject. The starting weld pass area is defined as 19 mm (3/4 inch) maximum on each side of the start-stop weld center line. Should any sample for any of the above tests fail to comply with minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

4.4.5.2 Rotating band welder reduced sampling. Sampling of one in each half shift (not to exceed five hours) of production, or fraction thereof, per welding machine in operation may be implemented after a welding machine has been qualified. Each sample shall have a band chemical analysis taken in accordance with 4.4.5.1 a. One sample for each welding machine in operation shall have the band chemically removed and examined in accordance with 4.4.5.1 b. The body having the highest acceptable iron content rotating band during each shift of post

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qualification sampling, when tested per 4.4.5.1 a., shall be tested in accordance with 4.4.5.1 c. Failure of the body to comply with the specified requirements will require rejection of all bodies having bands welded by the same welder during the same sampling period. Failure of any body will also require that the body exhibiting the highest iron content from each welder will be similarly tested. Failure of any body will result in rejection of the quantity from each discrete welder represented by the sample. Two consecutive failures on any welding machine will mandate the return to the original welder qualification sampling of 4.4.5. However, post qualification sampling may be implemented after the first five banded bodies have been found acceptable.

4.4.5.3 Magnetic particle inspection (Rotating band area). The inspection shall be in accordance with ASTM E1444, using the continuous, wet fluorescent method. The suspension shall be applied to the machined area beneath the rotating band. The magnetic solid content of the suspension shall be maintained at 0.2 - 0.4 milliliters (ml) of solids per 100 mls of suspension when oil is used and 0.15 - 0.35 mls of solids per 100 mls suspension when water is used. After draining the excess suspension, the rotating band area shall be inspected under ultra-violet light for the presence of defects. The adequacy of the inspection process shall be checked at the start of the daily production and at eight-hour intervals thereafter using a standard in accordance with dwg.9280441. The plug in the standard shall be clearly defined for a full 360 degrees on the outside of the body. If the inspection process is determined to be inadequate, the condition corrected and the proceeding 8 hours production shall be inspected. The requirement of ASTM E 1444 to check system effectiveness by means of steel ring specimen shall be waived. The provisions for the removal of additional material shall not apply to the inspection beneath the rotating band on the samples selected in accordance with 4.4.5.

4.4.5.4 Rotating band stress relief verification. A sample from the two highest iron content first article chemical analysis samples shall be used to verify stress relief of the heat affected zone (if applicable). The heat affected zone specimen shall be taken from location R on Figure 2. Rockwell hardness measurements shall be taken starting 1.59 mm (1/16 inch) from the band seat center and in 1.59 mm (1/16 inch) increments moving toward the interior of the projectile to a depth of one half inch below the band seat. Testing shall be in accordance with ASTM E18. Failure to meet the requirement of

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3.8.5 shall be cause for rejection. The stress relief shall be verified each time a coil is replaced or a process change is made.

4.4.6 Hydrostatic test. The unpainted projectile shall be mounted in a suitable test fixture and be subjected to the pressure specified on the applicable drawing. Any projectile which fails to comply with the applicable requirements shall be classed defective and removed from the lot.

4.4.7 Ultrasonic inspection. The projectiles shall be ultrasonically inspected using equipment and procedures in accordance with 4.4.7.1. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. Standards, in accordance with 4.4.7.3, shall be used to check the effectiveness of the inspection process. Any projectile failing to meet the requirements of 3.11 shall be classed defective and removed from the lot. All projectiles rejected by ultrasonic inspection shall be visually inspected at the location of the ultrasonic indication. An ultrasonic critical defect shall be defined as any ultrasonic reject that has a visual crack or pipe or a visual defect other than a crack or pipe (lap, fold, seam, etc.) that has an ultrasonic signal that is twice the level of the ultrasonic standard notch for that area. Visual defects other than cracks or piping shall be statically evaluated by a level II inspector to determine if a critical defect exists. Critical defect determination shall be accomplished before the start of the next production shift.

4.4.7.1 Ultrasonic test equipment and procedures. The ultrasonic inspection equipment and procedures shall meet the requirements of drawing 9280336. The ultrasonic inspection shall be performed with the use of a liquid to assure a reliable coupling to transmit ultrasonic energy. Should a stream or thin film of liquid be used to couple this energy, controls shall be used to monitor the coupling and assure its effectiveness. Such monitoring is not required if submergence in a liquid is used. Bodies shall be inspected using ultrasonic energy at right angles to assure the orientation does not prevent detection of discontinuity defects/flaws and accurate measurement of the wall thickness and variation. Using ultrasonic energy at right angles requires at least two separate sources of ultrasonic energy for defect/flaw inspection. This means that in the wall and bourrelet ultrasonic regions, ultrasonic energy shall move within the walls of the projectile with separate significant axial or circumferential components of velocity to detect flaws

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located on the outside, inside or within the walls. For the base region, significant ultrasonic velocity components shall separately be radial and circumferential. Criteria for detection of cracks are machined notches which simulate cracks. Details are contained on drawings 9280308 and 9280309. All notches machined on the standards are targets for rejection. That is, any projectile that sends back an echo signal that is greater than or equal to the reject level established in drawing 9280336 will be considered a rejectable defect. Relief portions of radially oriented notches where the full depth approaches the surface are not inspection criteria. A third source and direction of ultrasonic energy is necessary to accurately measure wall thickness and variation. Criteria for detection of wall thickness and variation are holes and machined bands. Details are contained on drawing see 13066810. All holes and machined bands on the standard are targets for rejection. That is, any projectile that is measured as having a wall thickness equal to or below the minimum level, or a difference in wall thickness equal to or greater than the allowable maximum variation level, shall be rejected.

4.4.7.2 Calibration and operation procedures and equipment. The adequacy of the inspection process shall be checked by using the standard at the start and finish of each production shift or change in operator, as well as at the shift midpoint during production. All channels shall properly indicate a rejection signal when encountering the notches mentioned in 3.11. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected. Equipment used shall provide for a positive activation of a buzzer or alarm as well as a light system for each channel to alert the operator of the signal indications that are the criteria for rejection.

4.4.7.3 Ultrasonic standards. The ultrasonic inspection standard shall be provided by the contractor for use during the ultrasonic inspection. The standards shall be machined from an acceptable projectile in accordance with drawings 9280308, 9280309, and 13066810, and shall be inspected for all drawing requirements to determine their acceptability for use. Government verification of all inspections performed on the standard is required.

4.4.8 Protective coating - thickness. Two samples shall be selected at random to represent each shift or fraction thereof, of continuous paint production. The sample projectiles shall be

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inspected for paint coating thickness at a minimum of four separate locations on each projectile. Any projectile failing to meet the applicable requirements shall be classed defective and every projectile painted since the last acceptable sample shall be rejected.

4.4.9 Salt spray. After qualification in accordance with TT-C-490 or commercial equivalent as approved by the contracting officer, two(2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.9.1 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by ASTM B117 and TT-C-490 or commercial equivalent as approved by the procuring contracting officer, with the exception that test samples shall be selected in accordance with 4.4.9.

4.4.10 Ballistic performance test for structural integrity. These tests shall be performed at a Government proving ground or Government approved installation in accordance with the applicable acceptance test procedure. The sample projectiles shall be inert loaded, assembled and fired at ambient temperature in a howitzer using a PXR6297A1 propelling charge. The propelling charges shall be uniformly temperature conditioned to provide a chamber pressure of 399.9 + 27.6 Megapascals (Mpa) (58,000 + 4000 PSI). Photographic and instrumental coverage shall be provided to assure observation of round at muzzle exit and over the predicted trajectory. Range and deflection shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts.

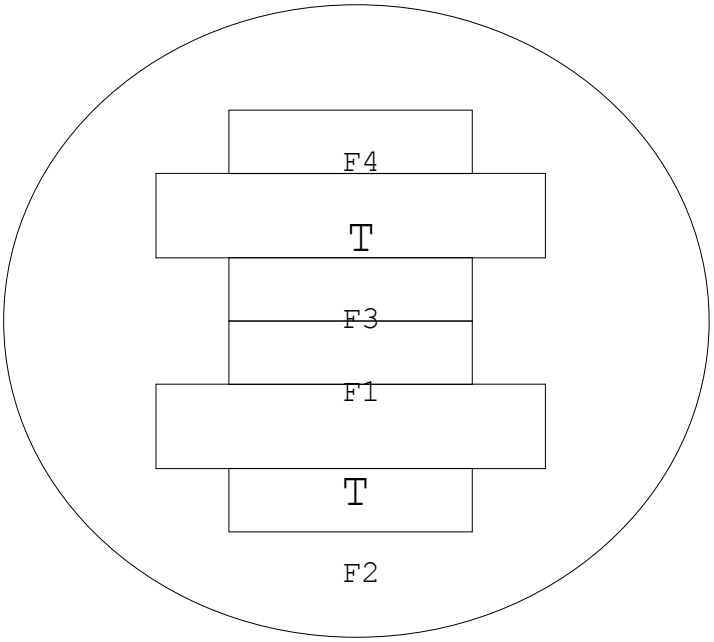
4.4.10.1 Initial inspection. Beginning with the first lot produced and continuing until three consecutive lots have complied with requirements specified, 20 sample projectiles shall be randomly selected for this test, 20 sample projectiles shall be observed for structural integrity of parts when tested

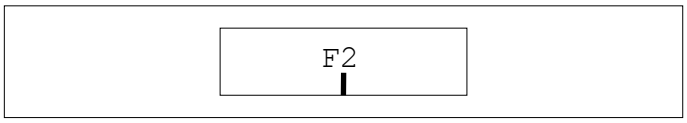
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in accordance with 4.4.10. If any sample projectiles fail to comply with the requirements for structural integrity of parts, the lot shall be rejected.

4.4.10.2 Reduced inspection. After three consecutive lots have complied with 4.4.10.1, 10 sample projectiles shall be randomly selected from each lot for this test. The 10 sample projectiles shall be observed for structural integrity of parts. If any of the sample projectiles fail to comply with the requirement for structural integrity of parts, the lot shall be rejected. If the lot is rejected the inspection criteria reverts back to paragraph 4.4.10.1.

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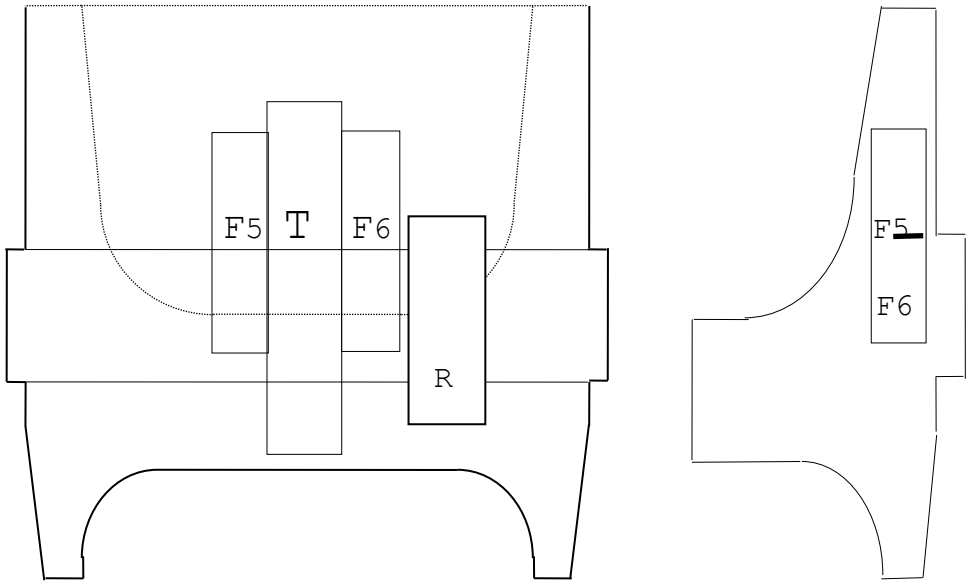
Approximate location of base tensile
and fracture toughness specimens
(T = Tensile F = Fracture Toughness)

Recommended base fracture toughness sample size:

Thickness B= 9.525 mm (.375 inch)
Width W= 19.05 mm (.75 inch)
Span S= 80.01 mm (3.15 inch)

Figure 1. Projectile base mechanical properties test locations

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Approximate location of side wall tensile
and fracture toughness specimens.
Fracture toughness specimens may be taken
90 degrees from tensile specimens.

(T = Tensile F = Fracture Toughness R = Stress relief)

Recommended side wall fracture toughness sample size:

Thickness B= 6.604 mm (.26 inch)
Width W= 13.208 mm (.52 inch)
Span S= 54.560 mm (2.15 inch)

Figure 2. Projectile rear mechanical properties test locations

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The projectile covered by this specification is intended for use on the Projectile, 155MM, HE, M795. This specification contains requirements suitable for production.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Requirements for the following. The chemical composition of the body steel should be certified. Independent verification of mechanical properties should be conducted in accordance with this specification. No projectile should be reheat treated without prior Government procuring contracting officer approval.

c. For acquisition purposes, the contract or order should specify packaging requirements. The projectiles should be palletized and marked in accordance with drawing 12944446 or alternative 9329574 and be ordered by the applicable stock number.

d. Requirements for submission of first article sample.

e. Submission of contractor acceptance inspection equipment. (See 6.3)

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f. Provisions for Certification of nondestructive test (NDT) personnel. (See 6.6)

g. Establishment of visual defect standards. (See 6.4)

h. Submission of rotating band procedures. (See 6.5.)

j. Submission of ballistic test data and ammunition data cards. (See 6.8)

k. Provisions for critical characteristic controls. (See 6.9)

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Prior to initiation of ultrasonic inspection, the contractor should submit, as part of the AIE submittal, for written approval one copy of his proposed procedures, including descriptions of equipment (contractor design) and operation procedures. No change to the procedure should be made after the original approval without submission and approval of the proposed changes.

6.4 Visual standards. Visual standards for evaluation of metal defects will be established for each contract after the first month of production. However, establishment of visual standards will not waive the requirements for compliance with 3.6.1

6.5 Rotating band application

6.5.1 Inspection of band seat.

6.5.1.1 Rotating band seat location and runout. The contractor may use any control method that will assure the band seat meets the dimensional requirements of the applicable drawing. The inspection plan must include all the required criteria to assure the dimensional aspects of the drawing will be equal to the characteristics defined in 4.3.2.7. When the contractor establishes a control method, the manufacturing procedure, and inspection plan should be forwarded to the Technical Agency as part of the AIE submittal prior to use in accordance with 3.8.2. No change to these controls will be made once established without approval from the contracting officer.

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6.5.2 Submission of rotating band welding procedures. The procedures for applying the rotating band and for maintaining, controlling and monitoring the rotating band application should be submitted to: Commander, TACOM-ARDEC, ATTN:AMSTA-AR-WEA-R, Picatinny Arsenal, New Jersey 07806-5000 for review and concurrence. It is recommended that the welding operation of the overlay welded band be guided by Rotating Band Welding Procedure for M795 drawing 12982985.

6.5.2.1 Submission of rotating band swaging procedures. The contractor and the government should agree upon swaging machine controls which have been shown to produce acceptable rotating bands. The manufacturing procedure and the inspection plan for controlling the process should be part of the AIE submittal prior to implementing into production. Once agreed upon these controls should not be changed without prior Government approval. The procedures for applying the rotating band and for maintaining, controlling and monitoring the rotating band application should be submitted to: Commander, TACOM-ARDEC, ATTN:AMSTA-AR-WEA-R, Picatinny Arsenal, New Jersey 07806-5000 for review and concurrence.

6.5.2.2 Submission of alternative nondestructive test for swaged band diametrical clearance. The contractor may submit as an alternative inspection method an ultrasonic test procedure including a calibration standard and ultrasonic test equipment. The inspection of the rotating band should cover the aft end rotating band if the entire band cannot be inspected.

6.5.3 Welded rotating band application. The rotating band should be overlay welded using the gas metal arc welding process in accordance with the applicable drawings and specifications. Gas metal arc welding process, in this instance, is defined as the use of a consumable electrode with a non-electrode auxiliary wire. The width of the band should be obtained by oscillating the welding head and the auxiliary wire feed system.

6.5.4 Suggested etching procedures for removing gilding metal overlay. Immerse body in commercial copper stripper which is cyanide and chromate free and prepared in accordance with manufacturer's directions. In a fresh solution, an overlay layer 0.254 to 0.381 mm (.010 to .015 inch) thick will dissolve in 10 to 14 hours. Renew solution as necessary.

6.6 Personnel qualification and certification for non-destructive testing (NDT). Personnel performing the ultrasonic and magnetic particle testing should be qualified and certified

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in accordance with recommended practice NAS 410 and procedures approved by ARDEC. Minimum qualification levels should be as follows:

Level III: Review and approval of NDT procedures, equipment and personnel certification plus the following;

Level II: Set-up, calibration, modification of NDT equipment and procedures plus the following;

Level I: Operation of NDT equipment.

A log book shall be maintained by the operator of each system which as a minimum, documents the use of standards, transducers, instrumentation, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

6.7 Process notes. The following notes are provided for information only:

a. Heat treat - Austenitize – 802 °C to 843 °C (1475 °F to 1550 °F) for one hour minimum.

- Oil Quench - 43 °C to 60 °C (110 °F to 140 °F)

- Temper - 560 °C (1040 °F) min. for 1 hr. minimum

b. Cooling of forgings should not be retarded. Slow cooling can cause the formation of carbides.

c. Spheroidizing after forging can be beneficial in rough machining, nosing and heat treatment.

d. Quenching over spuds is recommended to assure adequate quenching of the base.

e. The hydro test plug should be designed with a one quarter inch, minimum thread relief to prevent loading of initial threads on the projectile.

f. Examination should be made of all hydro test plugs after a projectile is blown during testing. Badly nicked or

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damaged plugs may cause destruction of otherwise acceptable items.

6.8 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards should be forwarded to Commander, TACOM-ARDEC, ATTN: AMSTA-AR-QAA-R, and AMSTA-AR-FSA-M, Picatinny Arsenal, NJ 07806-5000, and AMSMC-QAD-I Army Master Depository, Rock Island, IL 61299-6000.

6.8.1 Range and deflection. Range and deflection are not technical requirements of this specification. They are mandatory performance requirements that are achieved by design and are affected by variability in the manufacturing process that may not be completely controlled by the present TDP.

6.9 Critical characteristic controls. All production contracts should contain requirements for handling critical defects. As part of all proposals, suppliers should be asked to describe policies, procedures and controls for all operations associated with safety characteristics, how they are documented and maintained under the supplier’s integrated management system.

6.10 Tool Control. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided that the contractor has established a correlation between the tool dimensions and the components dimensions prior to the start of production. The contractor should be responsible for maintaining adequate surveillance to assure the drawing requirements are being met throughout the life of the tools.

6.11 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.12 Packaging Drawings. Drawings listed in this section are for reference only. The government's current packaging for the M795 Metal Parts Assembly as defined by Technical Data Package (TDP) 9329574 and 12944446, is included for reference. If the contractor chooses to build to the current approved TDP (9329574 and 12944446) the Government will waive the testing required to meet the First Article performance test requirements

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of the packaging specification. The contractor may use the information contained in the TDP as guidance for its own design. Unless otherwise noted, the custodian of the following drawings is the U.S. Army Tank-Automated Command- Armament Research, Development and Engineering Center, Picatinny Arsenal, NJ 07806-5000;

6.13 Definitions

a. Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard which is used to assure conformance of material to the contract requirements.

b. Automated Acceptance Inspection Equipment (AAIE). Equipment for which no, or minimal, human involvement is required in the acceptance determination. This includes equipment employing probes/sensors/transducers that are automatically manipulated to perform measuring and detection functions.

c. Heat treatment. Heat treatment is understood to be any thermal process which brings about a modification in the mechanical and metallurgical properties of the metal involved.

d. Mult. The term "mult" is short for "Multiple", which is a piece of stock for forging that is cut from the bar to provide the material for an individual part.

6.14 Critical classification notes.

6.14.1 Hydrostatic test. Metal parts defects, except rotating band gaps, large enough to fail the hydrostatic test of the explosive loaded projectile are likely to cause premature functioning from break-up of the projectile in the gun during firing.

6.14.2 Ultrasonic test. Metal parts defects, except rotating band gaps, large enough to fail by ultrasonic test of the explosive loaded projectile are likely to cause premature functioning from break-up of the projectile in the gun during firing.

6.14.3 Magnetic particle test. Metal parts defects, except rotating band gaps, large and deep enough to fail by Magnetic particle test of the explosive loaded projectile are likely to

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cause premature functioning from break-up of the projectile in the gun during firing.

6.14.4 Ballistic lot acceptance test. Any premature projectile function or metal parts separation in the gun or in flight is likely to cause injury or death for the Soldier firing or supported by the howitzer.

6.14.5 Rotating band failure. Loss of a rotating band in flight or in bore may result in projectile instability and subsequently a short round potentially resulting in injury or death for Soldiers in the field.

6.14.6 Shearing of rotating band. A sheared rotating band may result in a failure of the projectile to fully spin up, resulting in projectile instability and subsequently a short round potentially resulting in injury or death for Soldiers in the field.

6.15 ARDEC specifications. Copies of ARDEC specification DTL9326055-1 may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.

6.16 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

6.17 Subject term (keyword) listing.

Hydrostatic test
Reduced inspection
Salt spray
Plane-Strain fracture toughness test
Ultrasonic test

Custodian:
Army – AR

Preparing Activity:
Army – AR
(Project 1320-2017-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.dla.mil.

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